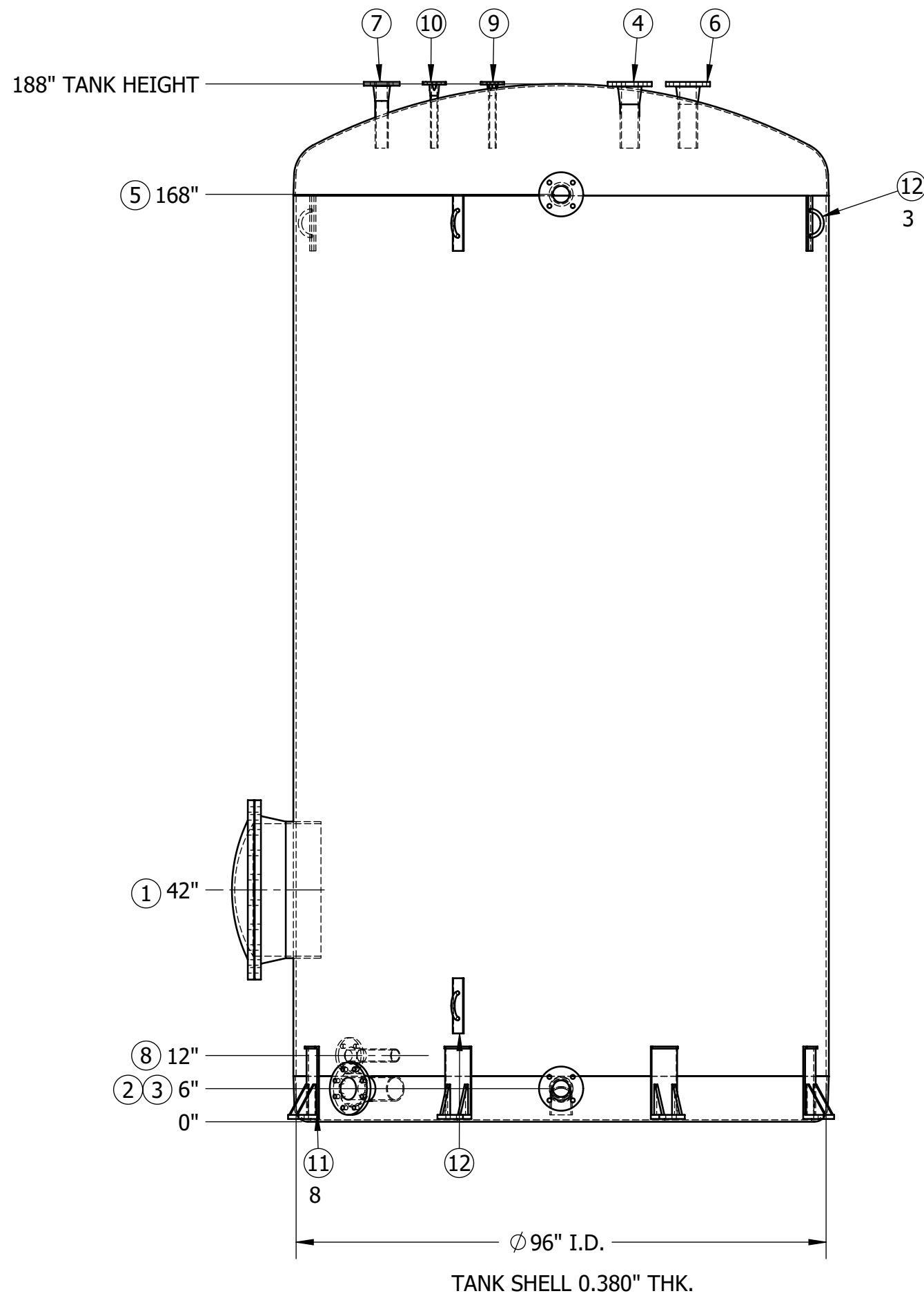
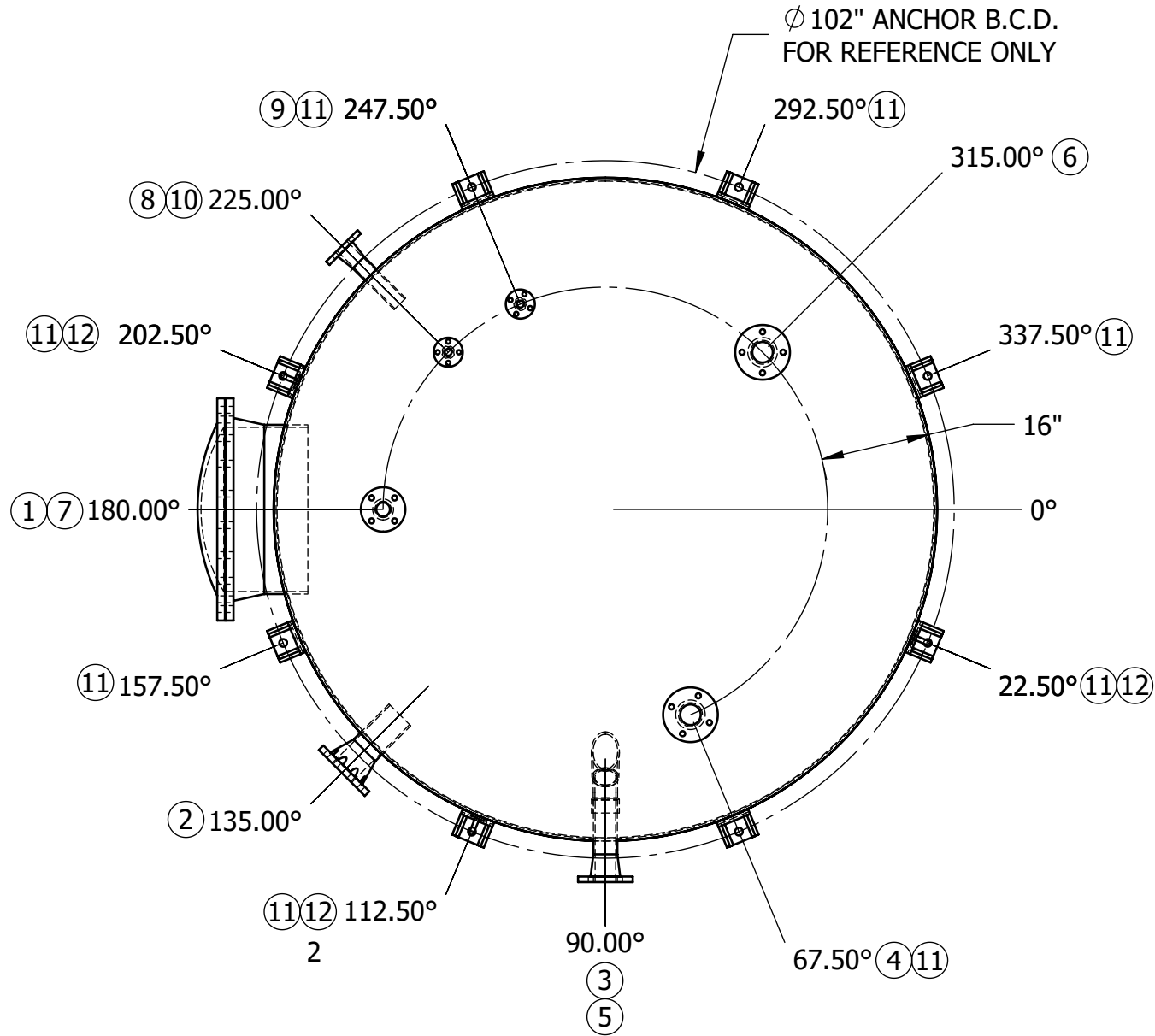




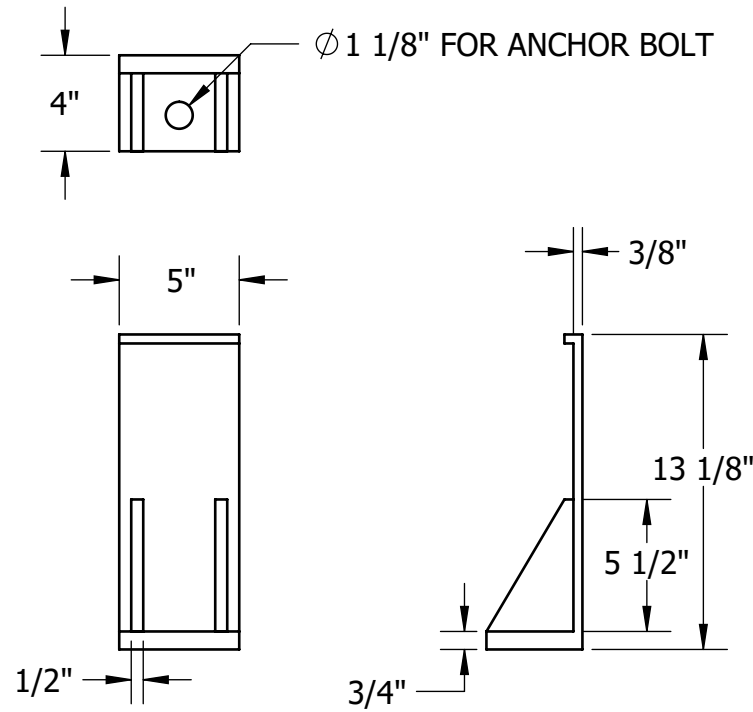
☐ APPROVED BY: _____

☐ APPROVED AS NOTED DATE: _____

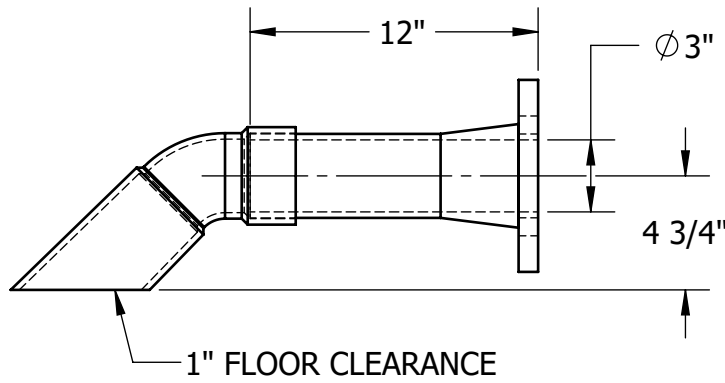
	Project No.: 22MH194			
	P.O. No.:	8027	Tag No.:	
1	Approved Without Comments			
2	Approved With Comments; Revise and Resubmit			
3	Revise and Resubmit			
4	For Information Only			
5	Certified as Final Data			

Julie Walker
Procurement/Vendor Data Contact

8/21/2024
Date



ITEM 11
6000 POUND HOLD DOWN 304 SS WITH HOLE
8 REQ'D



ITEM 3
3" BLADE GUSSETED FLANGE W/SIPHON ADAPTER

REVISION HISTORY			
REV.	DESCRIPTION	DATE	APPROVED
A	WIND WAS 140MPH, UPDATED ORIENTATIONS	12/7/2023	BJH
B	ITEM 4 WAS AT 31.5°, ITEM 6 WAS AT 45°, UPDATED TEMP., UPDATED SEISMIC AND ADDED HOLD DOWNS	1/30/2024	BJH
C	HOLD DOWN QNTY WAS 7	02/16/2024	ATB

SERIAL NUMBER	TOLERANCES - UNLESS OTHERWISE SPECIFIED +/- .1/4" ON DIMENSIONS	
24418	NOZZLE INSIDE DIA. UP TO 10 INCH 10 INCH & LARGER	PERMISSIBLE ANGULAR DEVIATION 1° 1/2°
S.O. 37161 FOR APPROVAL	PROPRIETARY NOTICE NOTICE: THIS DRAWING WAS ORIGINATED BY AND IS THE PROPERTY OF DESIGN TANKS LLC. WHO RESERVES ALL PATENT, PROPRIETARY DESIGN, USE, SALE, MANUFACTURING, AND/OR REPRODUCTION RIGHTS THERETO. UNDER PENALTY OF LAW MAY NOT BE USED IN ANY FORM BY ANY COMPANY WITHOUT THE EXPRESSED WRITTEN CONSENT OF DESIGN TANKS LLC.	

ITEM	QTY.	DESCRIPTION	PROJECTION	NOTES
1	1	24" 25 PSI SIDE MANWAY W/BLIND, INCLUDING VITON GASKET/316 SS BOLT SETS	7"	
2	1	4" BLADE GUSSETED FLANGE	6"	DRAIN
3	1	3" BLADE GUSSETED FLANGE W/SIPHON ADAPTER	6"	PUMP SUCTION
4	1	3" BLADE GUSSETED FLANGE	6"	LOOP RETURN INLET
5	1	3" BLADE GUSSETED FLANGE	6"	RO WATER INLET
6	1	3" BLADE GUSSETED FLANGE	6"	SPARE
7	1	2" BLADE GUSSETED FLANGE	6"	VENT
8	1	2" BLADE GUSSETED FLANGE	6"	LVL. TRANS.
9	1	1" BLADE GUSSETED FLANGE	6"	N2 SENSING
10	1	1" BLADE GUSSETED FLANGE	6"	N2 INLET
11	8	6000 POUND HOLD DOWN 304 SS WITH HOLE		
12	4	HEAVY DUTY 304 SS LIFT LUG		
13	1	CONTENT LABEL		
14	1	DESIGN TANKS LABEL		
15	1	NO AIR UNLOADING LABEL		
16	1	PE CALCS AND ANCHOR BOLT CALCS BY A TEXAS PE		

ORIENTATION ON THIS SHEET ARE TRUE ORIENTATIONS, ORIENTATION SHOWN ON SHEET 3 AND OTHER BOM ARE FOR QC PURPOSES ONLY.

TANK TO BE BUILT TO MEET **ASTM D3299** OR **ASTM D4097** ELEVATIONS FROM OUTSIDE BOTTOM. PROJECTIONS FROM OUTSIDE SHELL. ALL NOZZLE LOADS TO BE INDEPENDENTLY REACTED BY OTHERS. ALL NOZZLES ARE ANSI B16.5, 150# HOLE PATTERN, FLAT FACED UNLESS OTHERWISE NOTED. ALL FLANGED NOZZLES TO BE SPOT FACED. BOLT HOLES TO STRADDLE TANK CENTERLINE. BOLT TORQUE TO BE 30-35 FT.-LB. MAX. FLANGES WITHOUT BLINDS COVERED WITH HDPE COVERS PRIOR TO SHIPMENT. COUPLINGS SHIPPED WITH PVC PLUGS TO ENSURE NO CROSS THREADING. FASTENERS AND GASKETS FOR SERVICE UNLESS OTHERWISE NOTED. FASTENERS ARE TO BE 316 SS UNLESS OTHERWISE NOTED. IF REQUIRED NOZZLE CUTOUTS TO BE MARKED WITH NOZZLE MARK, S.O.# AND DATE. SEE BOM ON PRODUCTION DRAWING.

LINER: **ISOPHTHALIC POLYESTER/MEKP CURE, C-VEIL**
STRUCTURAL: **ISOPHTHALIC POLYESTER/MEKP CURE**
EXT. COATING WITH UV RESISTANCE:
SIDEWALL: **WHITE**
HEADS: **WHITE**
WAX USED AS REQUIRED

WARRANTY VOID IF DESIGN CONDITIONS ARE NOT SUPPLIED.

DESIGN CONDITIONS
CHEMICAL ENVIRONMENT:
ULTRA-PURE DEIONIZED WATER W/ RESISTIVITY OF 18MΩ
MAX. TEMP: **122° F**
PRESSURE: **12 Inches WC**
VACUUM: **5 Inches WC**
SPECIFIC GRAVITY: **1**
WIND: **147 MPH**
SEISMIC: **Sds=1.00, Sd1=0.60, Site Class=D, IMP Factor =1.0**
CAPACITY: **5672 GALLONS**
EMPTY WEIGHT: **1713 POUNDS**
FILLED W / CONTENTS WEIGHT: **48960 POUNDS**

EQUIPMENT NO.: **T-501**



DESIGN TANKS
SIOUX FALLS, SOUTH DAKOTA
www.designtanks.com

TITLE: FBDT-8'-0" DIA. X 15'-8" TOTAL TANK HT. 5,672 GALLON			
COLOR: WHITE	DATE: 11/20/2023	NO. OF TANKS: 1 DRAWN BY: BJH	
CUSTOMER: EVOQUA	SIZE DWG. NO. D 42302C		
RFQ: 112465	P.O. 4501843131	SHEET 1 OF 5	

STRUCTURAL RESIN:  HEAD CHOP GLASS: 

CHOP / HOOP FILAMENT WOUND FIBERGLASS FABRICATION NOTES

1.

LINER TO BE ISOPHTHALIC POLYESTER/C-VEIL WITH MEKP CURE 100 MIL THICK.
2.

STRUCTURAL TO BE ISOPHTHALIC POLYESTER WITH MEKP CURE, 0.07" PER LAYER.
3.

EXTERIOR WHITE COAT TO CONTAIN 3% STYWAX.
4.

FABRICATED TO APPLICABLE SECTIONS OF ASTM D3299.
5.

SOME ITEMS MAY BE ROTATED OR OMITTED IN ELEVATION VIEW FOR CLARITY.
SEE TOP VIEW FOR TRUE ORIENTATION.
6.

ELEVATIONS ARE FROM OUTSIDE BOTTOM.
7.

PROJECTIONS ARE FROM OUTSIDE SHELL.
8.

ALL NOZZLE LOADS TO BE INDEPENDENTLY REACTED BY OTHERS.
9.

ALL NOZZLES ARE ANSI B16.5 150# HOLE PATTERN, FLAT FACE UNLESS OTHERWISE NOTED.
10.

ALL FLANGED NOZZLES TO BE SPOT FACED.
11.

BOLT HOLES TO STRADDLE RADIAL CENTERLINES.
12.

BOLT TORQUE TO BE 30-35 FT.-LB. MAXIMUM.
13.

ALL FLANGES WITHOUT BLINDS TO BE COVERED WITH HDPE COVERS PRIOR TO SHIPMENT.
14.

FASTENERS AND GASKETS FOR SERVICE BY OTHERS UNLESS OTHERWISE NOTED.
15.

FASTENERS TO BE 316 SS UNLESS OTHERWISE NOTED.

LAMINATE SEQUENCE SYMBOLS

V - 10 MIL THICK "C" GLASS VEIL 7% GLASS CONTENT.
C - 45 MIL THICK 1.5 OZ./SQ. FT. CHOPPED "E" GLASS ROVING 30% GLASS CONTENT.
M - 38 MIL THICK 1.5 OZ./SQ. FT. CHOPPED "E" GLASS MAT 30% GLASS CONTENT.
R - 70 MIL THICK 28.3 OZ./SQ. YD. GLASS WOVEN ROVING 45% GLASS CONTENT.
N - 15 - 20 MIL THICK APERTURED NEXUS VEIL.
CH - 70 MIL THICK 2.4 OZ./SQ. FT. CHOPPED "E" GLASS ROVING AND 68 STRANDS OF 250 YIELD CONTINUOUS "E" GLASS HOOP STRANDS 50% GLASS CONTENT.

		Project No.: 22MH194	
	P.O. No.:	8027	Tag No.:
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5	Certified as Final Data		

Julie Walker

Procurement/Vendor Data Contact

8/21/2024

Date

REVISION HISTORY			
REV.	DESCRIPTION	DATE	APPROVED
A	WIND WAS 140MPH, UPDATED ORIENTATIONS	12/7/2023	BJH
B	ITEM 4 WAS AT 315°, ITEM 6 WAS AT 45°, UPDATED TEMP., UPDATED SEISMIC AND ADDED HOLD DOWNS	1/30/2024	BJH
C	HOLD DOWN QNTY WAS 7	02/16/2024	ATB

TOLERANCES - UNLESS OTHERWISE SPECIFIED	
+/- 1/4" ON DIMENSIONS	
NOZZLE INSIDE DIA. UP TO 10 INCH 10 INCH & LARGER	PERMISSIBLE ANGULAR DEVIATION 1° 1/2°
PROPRIETARY NOTICE	
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WARNING NOTES - FAILURE TO COMPLY WITH THE FOLLOWING NOTES WILL VOID WARRANTY

THIS TANK DESIGNED FOR ATMOSPHERIC PRESSURE ONLY (MUST BE VENTED)
ALL CLOSED TOP TANKS MUST HAVE A VENT OF EQUAL OR GREATER SIZE THAN
THE LARGEST INLET OR OUTLET.

VENT IS NOT DESIGNED FOR COMPRESSED AIR UNLOADING.

A COMBINATION VENT/OVERFLOW FLANGE INCREASES THE POTENTIAL FOR TANK
PRESSURIZATION IF THE OVERFLOW BECOMES CLOGGED.

FOUNDATION PAD MUST BE FABRICATED UNIFORM IN TEXTURE AND APPEARANCE, AND
MEET A SURFACE PLANE TOLERANCE (1/8" IN 10 FEET). PAD MUST BE DESIGNED FOR A
FULLY LOADED TANK, SOIL PROPERTIES AND FREEZE / THAW FORCES.
SPECIAL ATTENTION MUST BE GIVEN TO THE BOTTOM EDGE OF THE TANK. THIS EDGE
MUST BE IN COMPLETE CONTACT WITH THE PAD.

SHIMMING UNDER HOLD-DOWN LUGS (IF REQUIRED) SHOULD BE DONE WITH TANK
1/4 TO 1/3 FULL OF WATER. DO NOT TIGHTEN ANCHOR BOLTS WITHOUT SHIMMING
UNDER THE LUGS TO AVOID DAMAGING STRESSES ON THE TANK.

DESIGN CONDITIONS MUST BE PROVIDED TO CONFIRM CORROSION RESISTANCE OF RESIN SELECTION
WARRANTY VOID IF DESIGN CONDITIONS ARE NOT SUPPLIED.

DESIGN CONDITIONS

CHEMICAL ENVIRONMENT: ULTRA-PURE DEIONIZED WATER W/ RESISTIVITY OF 18MΩ
MAX. TEMP: 122° F
PRESSURE: 12 Inches WC
VACUUM: 5 Inches WC
SPECIFIC GRAVITY: 1
WIND: 147 MPH
SEISMIC: Sds=1.00, Sd1=0.60, Site Class=D, IMP Factor =1.0
CAPACITY: 5672 GALLONS
EMPTY WEIGHT: 1713 POUNDS
FILLED W / CONTENTS WEIGHT: 48960 POUNDS

EQUIPMENT NO.: T-501
SERIAL NUMBER: 24418

S.O. 37161
FOR APPROVAL

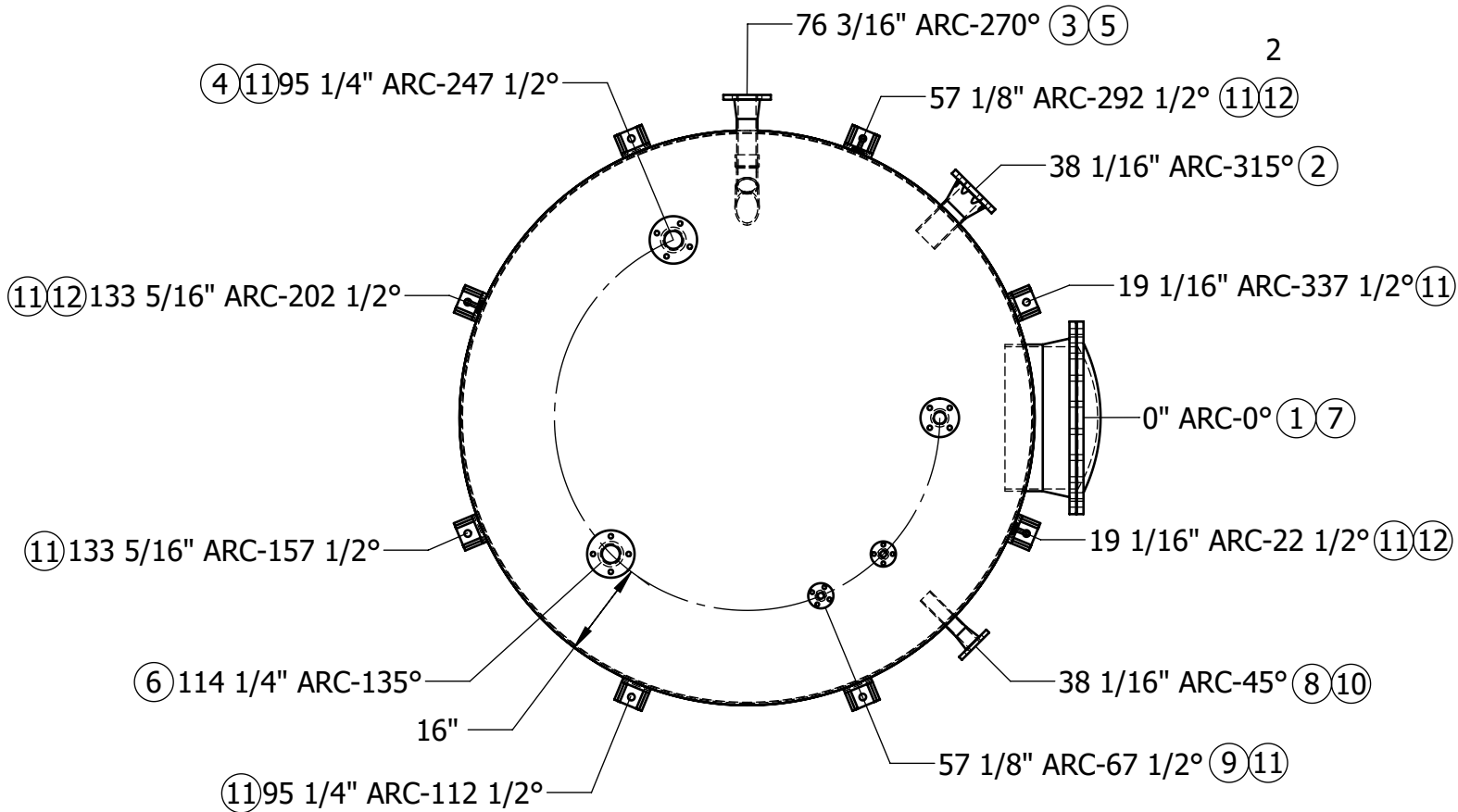
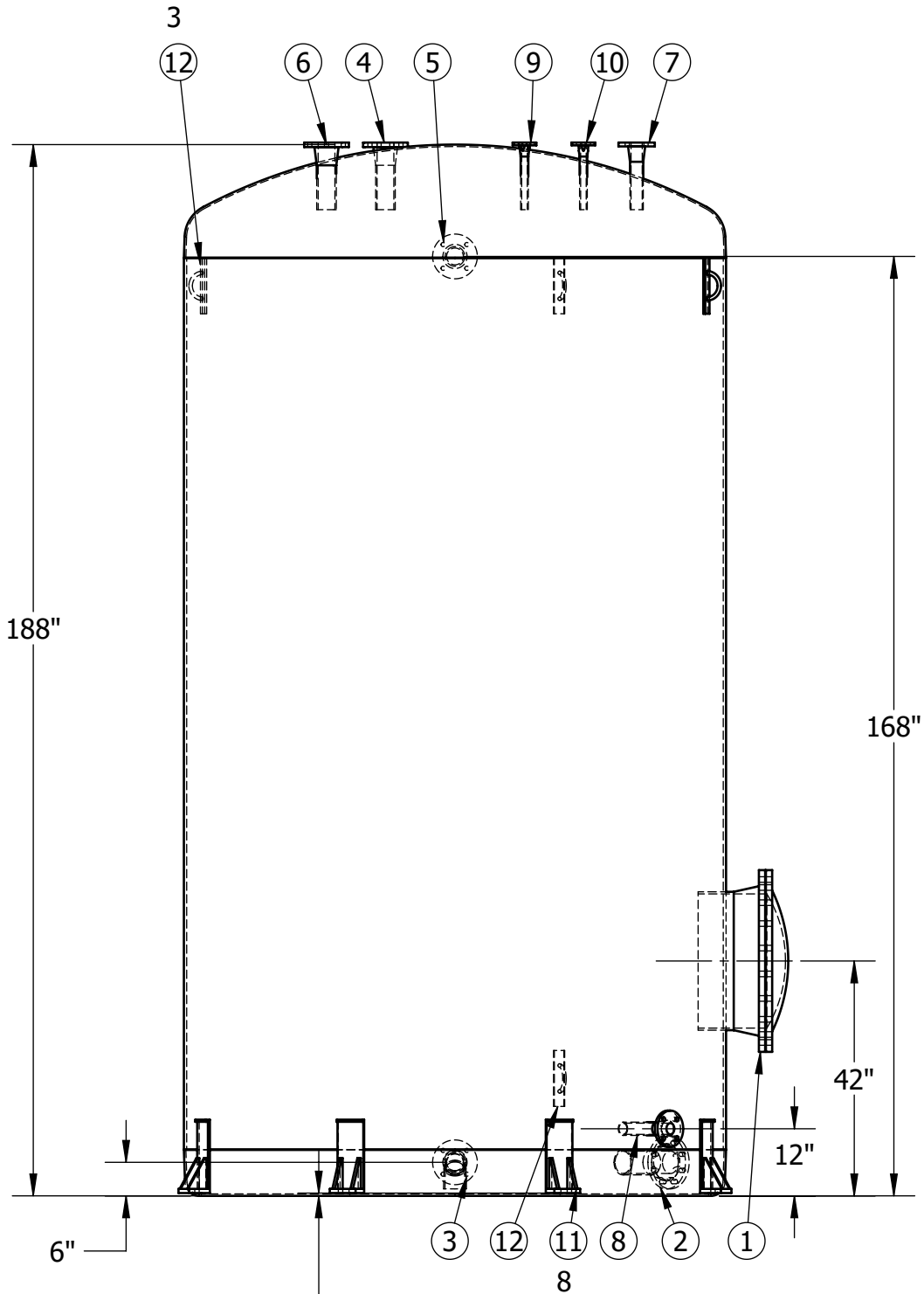
		SIOUX FALLS, SOUTH DAKOTA www.designtanks.com	
TITLE: FBDT-8'-0" DIA. X 15'-8" TOTAL TANK HT. 5,672 GALLON			
		NO. OF TANKS: 1	
COLOR: WHITE		DATE: 11/20/2023	DRAWN BY: BJH
CUSTOMER: EVOQUA		SIZE B	DRAWING NO. 42302C
RFQ: 112465		P.O. 4501843131	
		SHEET 2 OF 5	

HEAD CHOP GLASS:

STRUCTURAL RESIN:

SIDEWALL CHOP GLASS:

LINER RESIN:



	Project No.: 22MH194		
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Julie Walker

8/21/2024

Procurement/Vendor Data Contact

Date

TOLERANCES-UNLESS OTHERWISE SPECIFIED	
+/- 1/4" ON DIMENSIONS	
NOZZLE INSIDE DIA. UP TO 10 INCH 10 INCH & LARGER	PERMISSIBLE ANGULAR DEVIATION 1° 1/2°
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FOR APPROVAL



DESIGN
TANKS

SIoux FALLS, SOUTH DAKOTA
www.designtanks.com

TITLE: FBDT-8'-0" DIA. X 15'-8" TOTAL TANK HT. 5,672 GALLON

COLOR: WHITE

CUSTOMER:

EVOQUA

RFQ: 112465

P.O. 4501843131

NO. OF TANKS: 1

DATE: 11/20/2023 DRAWN BY: BJH

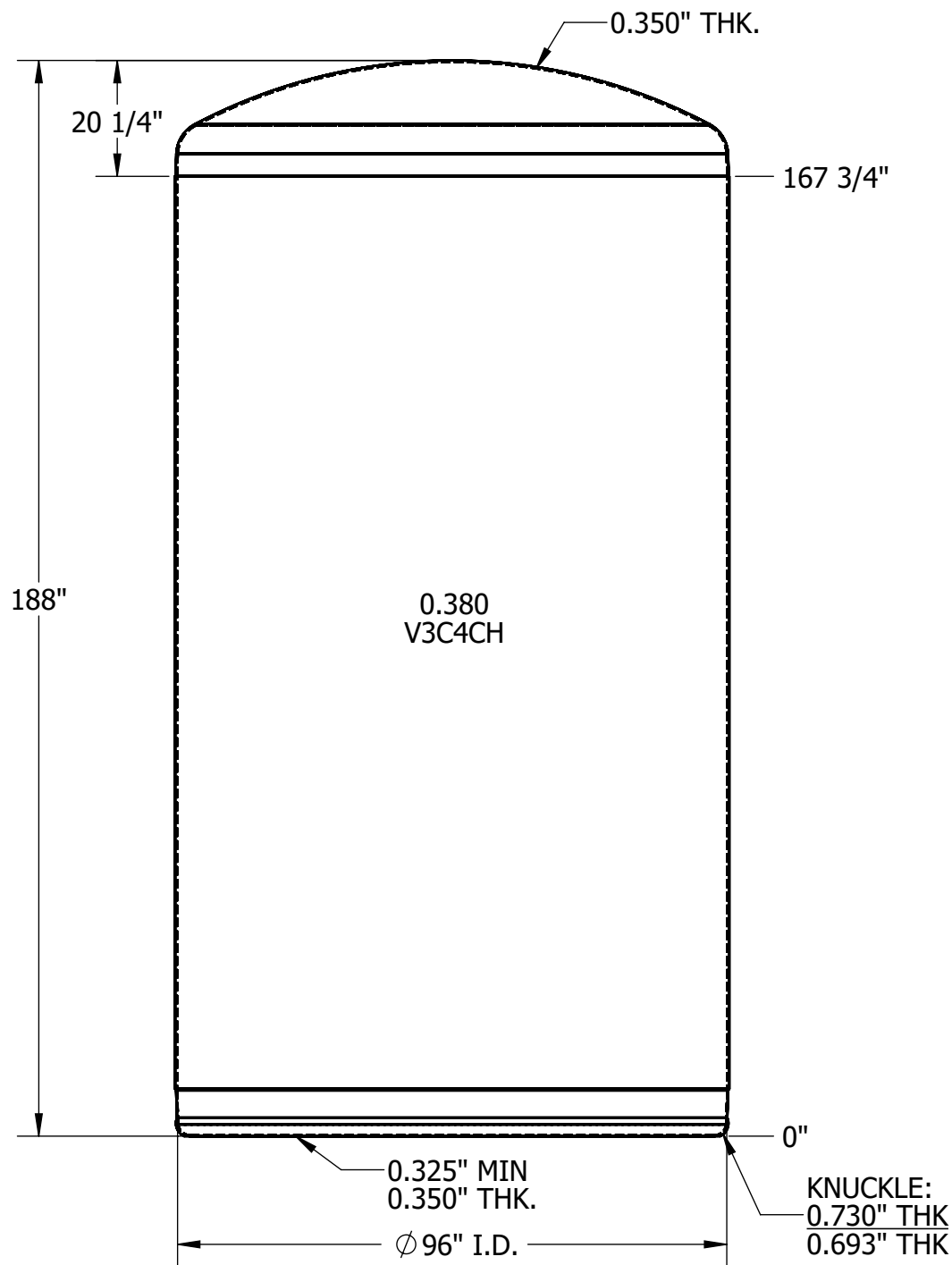
SIZE DRAWING NO.

B 42302C

SHEET 3 OF 5

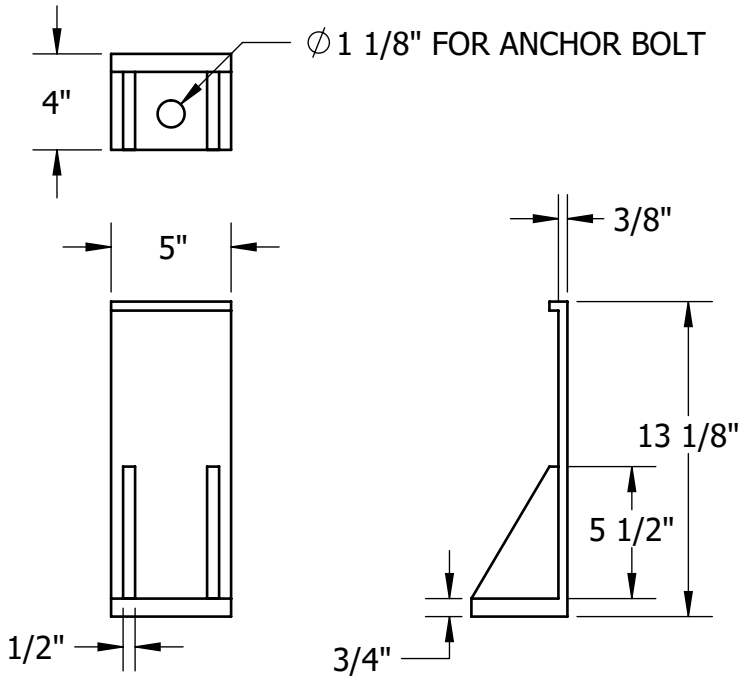
INTERIOR BONDING RESIN: ISOPHTHALIC POLYESTER WITH MEKP CURE
INTERIOR VEIL: C-VEIL
EXTERNAL BONDING RESIN: ISOPHTHALIC POLYESTER WITH MEKP CURE

STRUCTURAL RESIN: HEAD CHOP GLASS: SIDEWALL CHOP GLASS: LINER RESIN:

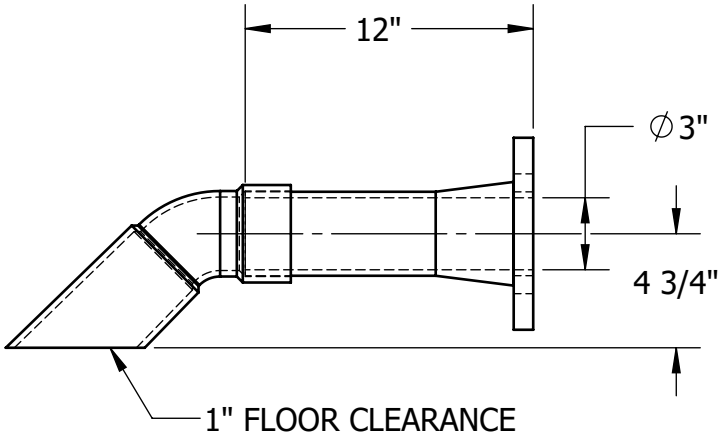


TANK CONSTRUCTION - REF. LAMINATE SEQUENCE ON SHT. 2

	Project No.: 22MH194		
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Julie Walker		8/21/2024	
Procurement/Vendor Data Contact		Date	



ITEM 11
6000 POUND HOLD DOWN 304 SS WITH HOLE 8 REQ'D



ITEM 3
3" BLADE GUSSETED FLANGE W/SIPHON ADAPTER

TOLERANCES-UNLESS OTHERWISE SPECIFIED	
+/- 1/4" ON DIMENSIONS	
NOZZLE INSIDE DIA. UP TO 10 INCH 10 INCH & LARGER	PERMISSIBLE ANGULAR DEVIATION 1° 1/2°
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FOR APPROVAL		DT DESIGN TANKS		SIOUX FALLS, SOUTH DAKOTA www.designtanks.com	
TITLE: FBDT-8'-0" DIA. X 15'-8" TOTAL TANK HT. 5,672 GALLON		NO. OF TANKS: 1		DATE: 11/20/2023 DRAWN BY: BJH	
COLOR: WHITE		CUSTOMER: EVOQUA		SIZE: B	
RFQ: 112465		P.O. 4501843131		DRAWING NO. 42302C	
				SHEET 4 OF 5	

INTERIOR BONDING RESIN: ISOPHTHALIC POLYESTER WITH MEKP CURE
INTERIOR VEIL: C-VEIL
EXTERNAL BONDING RESIN: ISOPHTHALIC POLYESTER WITH MEKP CURE

- 2 4 4 1 8

1 0 1 1 2 6 %

1 0 1 1 2 6 %

1 0 1 1 2 6 %

4 2 8 0 1 1 5

		Project No.: 22MH194	
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TOLERANCES-UNLESS OTHERWISE SPECIFIED

+/- 1/4" ON DIMENSIONS

NOZZLE INSIDE DIA. UP TO 10 INCH 10 INCH & LARGER

PERMISSIBLE ANGULAR DEVIATION 1° 1/2°

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FOR APPROVAL

 DESIGN TANKS

SIoux FALLS, SOUTH DAKOTA
www.designtanks.com

TITLE: FBDT-8'-0" DIA. X 15'-8" TOTAL TANK HT. 5,672 GALLON

NO. OF TANKS: 1

COLOR: WHITE

DATE: 11/20/2023

DRAWN BY: BJH

CUSTOMER: EVOQUA

SIZE: B

DRAWING NO.: 42302C

RFQ: 112465

P.O. 4501843131

SHEET 5 OF 5